

Trace results at end of test

On M-Series and T-Series equipment ForTest is possible to automatically get the test result trace at the end of test in a string format that can be decoded using the standard [communication protocol document](#).

Data can be taken from a PC , PLC or any controller type that can acquire data from a UART without the communication paradigm master-slave.

Please read the following example to understand where reach the right data

Example:

Program Number index 16 , length 5

:0101006191807190000100000012120000000000060020000000000000100000000000000300000000001000100000000000000030000000000005B

Test result index 27 , length 2

:0101006191807190000100000012120000000000060020000000000000100000000000000300000000001000100000000000000030000000000005B

This feature is available on different COM port depending from the Series of the equipment.

For more information please read the specific user manual of the equipment and the communication protocol document.

DATA MAPPING

Data received is a **117** bytes char array with following data structure:

Position	Length	Description
4	2	Hour
6	2	Minutes
8	2	Seconds
10	2	Day
12	2	Month
14	2	Year
16	5	Program Number
21	3	Linked programs
24	3	Kind of test



ForTest Italia s.r.l.

Sede legale:
Via Marsala nr. 17 - 41037 Mirandola (MO) - IT
Sede operativa:
Comparto Alfieri Maserati, Sez. 2A
Via Sibilla Aleramo, 7
41123 Modena (MO) - IT

P.I. / C.F. IT02934911203 REA MO-378687
Phone +39 059 557250
Fax +39 059 579022
www.fortest.com
info@fortest.com



27	2	Test results
29	2	Test status
31	10	Time remaining value
41	2	Time remaining unit measure
43	2	Time remaining decimal points
45	1	Pressure at beginning test phase sign
46	10	Pressure at beginning test phase value
56	2	Pressure at beginning test phase unit measure
58	2	Pressure at beginning test phase decimal points
60	1	Decay evaluated sign
61	10	Decay evaluated value
71	2	Decay evaluated unit measure
73	2	Decay evaluated decimal points
75	1	Auxiliary OUT1 sign
76	10	Auxiliary OUT1 value
86	2	Auxiliary OUT1 unit measure
88	2	Auxiliary OUT1 decimal points
90	1	Auxiliary OUT2 sign
91	10	Auxiliary OUT2 value
101	2	Auxiliary OUT2 unit measure
103	2	Auxiliary OUT2 decimal points
105	1	Temperature sign
106	5	Temperature value
111	2	Temperature unit measure
113	2	Temperature decimal points
115	2	Checksum

In new T-Series firmware there's an additional char array string of **176** bytes structured as follow:

Position	Length	Description
3	3	Kind of test
6	1	Filling pressure sign
7	5	Filling pressure value
12	2	Filling pressure unit measure
14	2	Filling pressure decimal points
16	5	Filling time set value
21	2	Filling time set unit measure
23	2	Filling time set decimal points
25	5	Settling time set value
30	2	Settling time set unit measure
32	2	Settling time set decimal points
34	5	Test time set value
39	2	Test time set unit measure

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41	2	Test time set decimal points
43	1	Max decay set sign
44	5	Max decay set value
49	2	Max decay set unit measure
51	2	Max decay set decimal points
53	1	N/D
54	5	Volume piece set value
59	2	Volume piece set unit measure
61	2	Volume piece set decimal points
63	3	Barcode length
66	40	Bar code data
106	17	Program name
123	17	Company name
140	17	Operator Code
157	17	Batch code
174	2	Checksum



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Unit measure

Value	Description
00	mbar
01	bar
02	hPa
03	Pa
04	psi
20	mbar/s
21	bar/s
22	hPa/s
23	Pa/s
24	psi/s
40	cc/h
41	cc/min
42	l/h
43	l/min
60	s
61	min
70	cc
71	l
80	--
81	%
82	bps
83	C°
84	conv/s
85	prg
86	chin
87	chout
88	Volt



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Test result

Value	Description
00	NO RESULT
01	OK
02	NOT OK
03	OK WITH RESERVE
04	INVERSE DECAY
05	NOT OK REFERENCE
06	NOT OK BELL TEST
07	NOT OK FLOW UNDER THRESHOLD
08	NOT OK PRESSURE OVER SCALE
09	NOT OK DECAY OVER SCALE
10	NOT OK PRESSURE UNDER TOLERANCE
11	NOT OK PRESSURE OVER TOLERANCE
12	NOT OK PRESSURE NOT REACH LEVEL
13	ABORT
14	NOT OK FLOW OVER THRESHOLD
98	ABORT
99	TEST IN PROGRESS



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Test status

Value	Description
01	Start
02	Delay on start
03	Pre-filling
04	Pre-filling
05	Pre-filling
06	Pre-filling
07	Discharge Pre-filling
08	Discharge Pre-filling
09	Discharge Pre-filling
10	Discharge Pre-filling
11	Filling
12	Filling
13	Filling Retry
14	Settling
15	Settling
16	Settling
17	Settling
18	Settling
19	Internal use
20	Internal use
21	Internal use
22	Internal use
23	Internal use
24	Internal use
25	Internal use
26	Test
27	Internal use
28	Internal use
29	Internal use
30	Obstruction test
31	Internal use
32	Internal use
33	Internal use
34	Internal use
35	Internal use
36	Bell Test
37	Internal use
38	Internal use



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39	Internal use
40	Internal use
41	Internal use
42	Internal use
43	Volume
44	Internal use
45	Internal use
46	Internal use
47	Internal use
48	Internal use
49	Test with OK result
50	Internal use
51	Internal use
52	Internal use
53	Internal use
54	Internal use
55	Internal use
56	Internal use
57	Internal use
58	Internal use
59	Internal use

Linked program

Value	Description
000	No linked program
00L	Linked program
00E	End of link sequence
00S	Stop of a linked sequence by abort



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Kind of test

Value	Description
000	Leak Test
001	Obstruction
002	Functional Cycle
003	Hydrogen
006	Differential
007	Absolute Test
009	Zero Center
100	Flow
101	Ramp+
102	Ramp-
103	Constant Flow



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DEVICE SETTINGS AND CONNECTION

T8000 Series

COM PORTS SETTINGS		▲	▼
SER0/USB modality:	Trace	[-----]	
SER0/USB Baud Rate:	9600	[bps]	
SER1 modality:	Nothing	[-----]	
SER1 Baud Rate:	9600	[bps]	
05/Feb/2017	08:49:35		
			

COM1 port pinout

COM1	
Pin	Descrizione
1	CANH (OPT)
2	TXRS232 SER1
3	RX RS232 SER1
4	CANL (OPT)
5	GND
6	TX RS232 SER0
7	RX RS232 SER0
8	RS485 B\
9	RS485 A

SER0 : UART communication
 ForTest protocol, Modbus RTU,
 Trace end of test
SER1 : Barcode, Printer

M4000 Series

COM PORTS ▲▼		
C1:	PC	[----]
Br1:	9600	[bps]
C2/usb:	Trace	[----]
Br2/usb:	9600	[bps]

Trace is available on COM2 of the equipment

To enable/disable this parameter from COM2 serial communication protocol is necessary to use the following commands:

":01C0200000000370000619" **ENABLE TRACE END OF TEST ON COM2**

":01C020000000037000001F" **DISABLE TRACE END OF TEST ON COM2**

M4000 COM2 pinout

Pin	Descrizione Mod. Programmazione	Descrizione Mod. Normale	Tipo di segnale
1	OUT 0 SCARICO	OUT 0 SCARICO	Output (+24 Vdc)
2	OUT 1 TEST	OUT 1 TEST	Output (+24 Vdc)
3	OUT 2 REJECT	OUT 2 REJECT	Output (+24 Vdc)
4	OUT 3 GOOD	OUT 3 GOOD	Output (+24 Vdc)
5	OUT 4	OUT 4	Output (+24 Vdc)
6	OUT 5	OUT 5	Output (+24 Vdc)
7	OUT 6	OUT 6	Output (+24 Vdc)
8	OUT 7	OUT 7	Output (+24 Vdc)
9	OUT 8	OUT 8	Output (+24 Vdc)
10	Tasto - / Abort	ING 0 ABORT	Input (Active at +24 Vdc)
11	Tasto + / Start	ING 1 START	Input (Active at +24 Vdc)
12	Tasto Enter	ING 2 START1	Input (Active at +24 Vdc)
13	Tasto TST	ING 3 START2	Input (Active at +24 Vdc)
14	Tasto SET	ING 4 START3	Input (Active at +24 Vdc)
15	---	ING 5	Input (Active at +24 Vdc)
16	TX0 RS232	TX0 RS232	Communication COM1
17	RX0 RS232	RX0 RS232	Communication COM1
18	TX1 RS232	TX1 RS232	Communication COM2
19	RX1 RS232	RX1 RS232	Communication COM2
20	RS485	RS485	Communication RS485
21	RS485	RS485	Communication RS485
22	24 Vdc Power Supply	24 Vdc Power Supply	+24 Vdc
23	24 Vdc Power Supply	24 Vdc Power Supply	+24 Vdc
24	GND	GND	Gnd
25	GND	GND	Gnd
26	GND	GND	Gnd

T6000 Series

PORTE COM		▲ ▼
C1-usb:	Trace	[---]
B1-usb:	9600	[bps]
C2:	Niente	[---]
B2:	9600	[bps]
16:23:46 21/October/2016		 

COM1 port pinout

COM1	
Pin	Descrizione
1	-
2	TX (RS232) – C2
3	RX (RS232) – C2
4	-
5	GND
6	TX (RS232) – C1
7	RX (RS232) – C1
8	-
9	-

C1 : UART communication
ForTest protocol , Modbus RTU,
Trace end of test
C2 : Barcode, Printer

M6000/8000 Series

COM PORTS SETTINGS ▲▼		
COM1 modality:	PC directly	[----]
COM1 Baud Rate:	9600	[----]
COM2/USB modality:	Trace end of test	[----]
COM2/USB Baud Rate:	9600	[----]

COM2 port pinout

COM2	
Pin	Descrizione
2	RX
3	TX
5	GND

COM2 : Barcode, Printer, Trace end of test